AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M140
Introducing Statistics

Midterm Examination (MTA)
Spring Semester 2024/2025
Date: 9 /April/2025

Number of Exam Pages: (3) (including this cover sheet)	**Time Allowed: 120 minutes**

Instructions:

Please read the following instructions before starting:

- Total Number of Questions:4
- This is a closed book exam.
- Be sure you write your name and ID on the External Answer booklet.
- This exam accounts to 30% of the total mark in the course.
- Budget your time according to the mark assigned to each question.
- Mobile phones and all other mobile communication equipment are NOT allowed.
- Calculators are allowed.

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Part 1: MULTIPLE CHOICE QUESTIONS

Solve all the questions. Each question is worth 1.5 marks, giving a total of 6 possible marks.

Q–1: What is the term used for the value that describes a sample drawn from a specific population?

a) parameter
b) statistic
c) variable
d) quantity

Q–2: The sample mean of the data 14, 20, 14, 18, 22, 24, 14, 16, 20 is

a) 20
b) 24
c) 18
d) 14

Q–3: If A and B are two independent events, with P (A)=0.3 and P (B)=0.6 then P(A∩B) is:

a) 0.9
b) 0.3
c) 0.18
d) 0

Q–4: Which percentile describes Q1:

a) 25th
b) 50th
c) 75th
d) 100th

Part 2: ESSAY QUESTIONS

Each question is worth 8 marks. You should answer all the three questions, giving a total of 24 possible marks.

Q–1: [3+1+1+3 Marks]

In a factory, a manager recorded the number of machines that required maintenance over six consecutive months. The recorded values are as follows:

9, 7, 11, 10, 13, and 7.

I. **Calculate the mean number of machines that required maintenance over these six months.**
II. **Find the median number of machines that required maintenance over these six months.**
III. **Find the mode of the number of machines that required maintenance over these six months.**
IV. **Calculate the sample standard deviation of the number of machines that required maintenance over these six months.**

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Q-2: [8 Marks]

An agricultural research organization tested a particular chemical pesticide to try to find out whether an increase in the amount of pesticide used would lead to a corresponding increase in crop yield. The following table shows the recorded values of crop yield in tons per hectare after using different amounts of pesticide in kilograms per hectare. Find the value of the correlation coefficient between the amount of pesticide used and the crop yield.

X Amount of pesticide (kg/ha)	1176	1065	1225	1181	1355	1263	1441
Y Crop yield (tons/ha)	2.9	4.7	8.9	15.2	16.8	24.2	26.0

Q–3: [2*4 =8 Marks]

A tech company wants to improve its training strategies. They conducted a survey of their full-time and part-time employees about their preferences for different training methods. The results are as follows:

Training Method	Prefer Lecture (In-Person)	Prefer Computer (Online)	No Preference	Total
Full-time	10	20	25	55
Part-time	12	25	8	45
Total	22	45	33	100

a) **If an employee is selected at random, what is the probability that he is a part-timer**

b) **If an employee is selected at random, what is the probability that he has no preferences?**

c) **If an employee is selected at random, what is the probability that he is a part-timer and have no preferences?**

d) **If an employee is selected at random, what is the probability that they are a part-timer GIVEN they Prefer Computer (Online)?**

End of questions

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